MIL-M-23045B(MC)
22 October 1968
SUPERSEDING
MIL-M-23045A(MC)
26 June 1963

MILITARY SPECIFICATION

METER ASSEMBLY, SKID MOUNTED

1. SCOPE

*1.1 This specification covers the requirements for one type skid mounted, portable, self-contained meter assembly.

2. APPLICABLE DOCUMENTS

*2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-A-200/8	- Aluminum Alloy Bar, Rod, Shapes, Tube and Wire, Extruded and Structural, Shapes, 6061.
QQ-A-250/11	- Aluminum Alloy, 6061, Plate and Sheet.
QQ-A-601	- Aluminum-Alloy Sand Castings.
QQ-P-416	- Plating, Cadmium (Electrodeposited).
RR-C-271	- Chains and Attachments, Welded, Weldless, and Roller Chain.
TT-E-527	- Enamel, Alkyd, Lustreless.
TT-S-735	- Standard Test Fluids; Hydrocarbon.
VV-F-800	- Fuel Oil, Diesel.
PPP-B-601	- Box, Wood, Cleated-Plywood.
PPP-B-621	- Box, Wood, Nailed and Lock-Corner.
PPP-T-60	- Tape: Pressure-Sensitive Adhesive, Water-proof, for Packaging.

MILITARY

MIL-B-121	- Barrier Material, Greaseproofed, Water-proofed, Flexible.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-G-3056	- Gasoline, Automotive, Combat.
MIL-C-3133	- Cellular Elastomeric Materials, Fabricated Parts.

FSC 4930

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MIL-G-5572	- Gasoline, Aviation, Grades 80/87, 100/130, 115/145.
MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-G-6183	- Cork and Rubber Composition Sheet, for Aromatic Fuel and Oil Resistant Gaskets.
MIL-W-8604	- Welding of Aluminum Alloys, Process for.
MIL-A-8625	- Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-L-21260	- Lubricating Oil, Internal-Combustion Engine Preservative.
MIL-C-27487	- Coupling Halve, Quick Disconnect-Cam-Locking Type.
MIL-V-58039	- Valves, Gate, Rising Stem, Double Acting, Aluminum.

STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing with Appropriate Test Methods.
MS33586	- Metal, Definition of Dissimilar.

DRAWINGS

MARINE CORPS

4121-2A	- Meter Assembly, Skid Mounted.
4121-2A-1	- Meter Assembly, Skid Mounted, Details.
4117-1	- Gasket, Synthetic Rubber and Cork.
4127	- Lead, Electrical Grounding.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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NATIONAL BUREAU OF STANDARDS Handbook H28 - Screw Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

*3.1 First article. Unless otherwise specified (see 6.2), as soon as practicable after award of the contract or order, and prior to submission of any complete item for acceptance, the contractor shall furnish a first article for determination of conformance to this specification. The first article shall be representative of the product which the manufacturer will subsequently produce in fulfillment of the contract. Approval of the first article by the activity concerned shall not relieve the contractor of his obligation to supply meter assemblies conforming to this specification. Any changes or deviations of the meter assemblies from the first article shall be subject to the approval of the contracting officer.

*3.2 Material. Material shall be as specified herein and on Drawing No. 4121-2A and 4121-2A-1. Material not definitely specified shall be submitted to the contracting officer for approval.

*3.2.1 Fuel resistance. All materials coming in contact with fuels shall be resistant to the fuels specified in TT-S-735, VV-F-800, MIL-G-3056, MIL-G-5572 and MIL-T-5624.

*3.2.2 Metals. Metal parts shall be corrosion-resistant or shall be protected against corrosion by means of chemical or electrolytic finishes and treatments, or a combination thereof. Steel components shall be cadmium plated in accordance with type II, class I of QQ-P-416.

*3.2.2.1 Dissimilar metals. Contact between dissimilar metals shall be avoided wherever practical. Where it is necessary that dissimilar metals be assembled in intimate contact with each other, an interposing insulating or mutually compatible material shall be used. Dissimilar metals are defined in MS33586. The following exceptions will be permitted:

(a) Bronze cam arms and bronze or stainless steel pins in the quick disconnect coupler fittings.

(b) Brass or bronze internal working parts.

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*3.2.2.2 Aluminum alloy sand castings. Aluminum alloy sand castings shall conform to class 3M, condition T-6 of QQ-A-601. Castings shall be of high quality, clean, sound and free from porosity, blow holes, cracks, and other production defects.

*3.2.2.3 Plate and sheet aluminum alloy. Plate and sheet aluminum alloy shall conform to QQ-A-250/11.

*3.2.2.4 Aluminum bars, rods, and shapes. Aluminum bars, rods, and shapes shall conform to QQ-A-200/8.

*3.2.2.5 Security chain. Chain for securing the dust caps and plugs to the associated fittings shall be in accordance with type II, class 3 of RR-C-271, and shall be equipped with two heavy duty bronze "S" hooks, (one at each end). The chain and the hooks shall be cadmium plated in accordance with QQ-P-416.

*3.2.2.6 Strainer screen. The strainer screen shall be Monel metal (or equivalent) (see 3.3.5).

*3.2.3 Gaskets. Flange gaskets shall be synthetic rubber and cork composition in accordance with class 1, medium grade, of MIL-G-6183 and Drawing No. 4117-1. Synthetic rubber gaskets in quick disconnect coupler fittings and dust caps shall be in accordance with MIL-C-27487. Spare parts chest gaskets shall be in accordance with type S, class SB-15 of MIL-C-3133.

*3.2.4 Fungus inert material. Unless otherwise specified herein materials which provide a nutrient medium for fungus and insects shall not be used.

*3.3 Design and construction. Each meter assembly shall be a completely self-contained, skid mounted unit, equipped with a meter, a register, one inlet, two outlets, a sediment strainer, a tool and spareparts chest and provisions for grounding.

*3.3.1 Meter. The meter shall be rotary type, capable of metering fuels at ranges between 20 and 350 United States standard gallons per minute (gpm) atdaamaximum pressure of 150 pounds per square inch (psi).

*3.3.1.1 Meter micrometer adjuster. Each meter shall be equipped with a micrometer adjuster for the purpose of calibrating the meter to the different fluids being metered. The adjuster shall be calibrated in terms of percentage dif-

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ferential between indicated volume and actual volume metered. The adjuster shall have a range from zero to five percent and shall be calibrated in increments of tenths of one percent.

*3.3.2 Valves. Each meter assembly shall be equipped with three 4-inch gate valves in accordance with MIL-V-58039, one at the inlet and one at each of the two outlets, as shown on Drawing No. 4121-2A.

*3.3.3 Register. A register shall be located on the top of the meter and shall be equipped with two counters.

*3.3.3.1 Upper counter. The upper counter shall be capable of registering from one tenth of one gallon up to 10,000 gallons and shall be capable of being reset to zero at any time without affecting the lower counter.

*3.3.3.2 Lower counter. The lower counter shall be cumulative type and shall register up to one million gallons.

*3.3.4 Tool and spare parts chest. A tool and spare parts chest shall be mounted upon the frame in front of the meter.

*3.3.4.1 Chest contents. The chest contents to be supplied with each meter assembly shall consist of the following parts:

(a) One strainer basket
(b) One strainer cover gasket
(c) One electrical ground wire.

*3.3.5 Sediment strainer. The sediment strainer shall be bucket type with a 50-mesh screen, and shall have a removable cover on the top for cleaning and a plug in the lowest point in the bottom for draining. The strainer shall be located between the inlet valve and the meter.

*3.3.6 Couplings. Each meter assembly shall be equipped with three quick disconnect coupling halves; one female conforming to type VIII of MIL-C-27487 at the inlet and one male conforming to type IV of MIL-C-27487 at each of the outlets as shown on Drawing No. 4121-2A. Dust caps and a dust plug conforming to types IX and X of MIL-C-27487, shall be provided with each appropriate fitting. Caps and plugs shall be secured with chains (see 3.2.2.5). All couplings shall provide a tight seal with mating fittings.

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*3.3.7 Skid and frame assembly. The skid and frame assembly shall be of all aluminum welded construction. The skids shall be a minimum of 3 inches wide, and shall be upturned 3-1/2 inches minimum at each end on a radius of 13 inches. A towing eye shall be provided in the end of each skid (four total) so that the meter assembly can be towed from either end; two lifting eyes shall be included in the frame, one at each end, and two sections of tubing with thumb screws shall be provided for securing the grounding rod when not in use. The skid and frame shall be in accordance with Drawing No. 4121-2A-1.

*3.3.8 Detachable parts. Each meter assembly shall be provided with a grounding rod, and a grounding lead with clips at each end for use during fueling operations, to minimize static electricity hazards. When not in use, the grounding lead should be stowed in the tool chest, and the grounding rod shall be stowed in brackets provided on the frame assembly. The ground lead shall conform to type I of Drawing No. 4127.

*3.3.9 Interchangeability. All parts having the same manufacturer's part number shall be physically and functionally interchangeable.

*3.3.10 Welding. All aluminum welding shall be by the inert gas shielded arc method in accordance with MIL-W-8604. The quality and strength of the welds shall be such as to enable the meter assemblies to meet the requirements of this specification.

*3.3.11 Special tools. No special tools shall be required for the operation and maintenance of the meter assembly or the components thereof.

*3.3.12 Screw threads. All screw threads shall be in accordance with H-28.

*3.3.13 Cleaning. All parts shall be clean and free from lint, sand, metal chips and weld splashes.

*3.4 Performance and product characteristics.

*3.4.1 Lifting. The meter assembly shall be capable of being hoisted by means of the lifting eyes without permanent deformation or damage to any part of the meter assembly when subjected to the test in 4.6.2.

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*3.4.2 Hydrostatic pressure. When tested as specified in 4.6.3, each meter assembly shall be capable of withstanding a hydrostatic pressure of 200 psi.

*3.4.3 Accuracy (metering). When tested as specified in 4.6.4, the meter assembly shall be capable of registering metered volume within 0.02 percent of true volume. Meter assemblies shall be capable of the above performance with any horizontal plane of the meter assembly inclined (in any direction), up to 5° from true horizontal.

*3.4.4 Towing. When tested as specified in 4.6.5, the meter assembly, completely filled with test fluid, shall be capable of being towed from either end by means of the towing eyes.

*3.5 Weight. The weight of each completely assembled meter assembly, dry, shall not exceed 270 pounds (lb), and shall be kept to a minimum consistent with the requirements of this specification.

*3.6 Finish.

*3.6.1 Anodizing. Aluminum parts not painted shall be color anodized in accordance with MIL-A-8625; the color shall match Marine Corps green, No. 23, lusterless (see 6.3).

*3.6.2 Painting. All items not anodized shall be painted to match Marine Corps green, No. 23, lusterless (see 6.3), in accordance with type A of MIL-T-704 except that the finished coat of paint shall conform to TT-E-527.

*3.7 Marking. Marking shall be in accordance with Drawing No. 4121-2A and MIL-STD-130.

*3.8 Nameplate. A U. S. nameplate shall be securely attached, in a conspicuous place, to each meter assembly. For the Marine Corps, these plates will be furnished, together with required data to be legibly imprinted in the blank spaces by the contractor, upon request to the Commandant of the Marine Corps, Code CSG, Washington, D. C. 20380.

*3.9 Workmanship. Each meter assembly shall be manufactured in a careful and workmanlike manner in accordance with sound practice consistent with the current state of the art.

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4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 General inspection and test requirements.

*4.2.1 Test conditions. Tests specified herein may be performed under any natural combination of prevailing ambient conditions. All test conditions shall be observed and recorded along with the test results for each test.

*4.2.2 Test fluid. Test fluid shall conform to TT-S-735. Water shall not be used for any test specified herein, including hydrostatic testing.

*4.3 Classification of tests. The inspection and testing of meters shall be classified as follows:

(a) First article tests (see 4.4 and 6.2).
(b) Acceptance tests (see 4.5).
(c) Preparation for delivery tests (see 4.7).

*4.4 First article tests. A completely fabricated meter assembly shall be subjected to all the inspections and tests specified in 4.6.1 through 4.7.3. These tests shall be made by the contractor in the presence of a Government inspector prior to any acceptance by the Government. These tests are usually performed once during the life of a contract and need not be repeated during the life of the contract unless a change is made in design, material or methods of construction.

*4.5 Acceptance tests. Each meter assembly submitted for acceptance shall be subjected to the inspections and tests specified in paragraphs 4.6.1 and 4.6.3. One of each 15 submitted for acceptance shall be submitted to the accuracy test as specified in 4.6.4. These tests shall be performed by the contractor in the presence of a Government inspector. Failure to pass any test shall be cause for rejection.

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4.6 Test procedures. Following are detailed examinations and test procedures by which conformance to the product requirements of this specification shall be determined:

4.6.1 Visual and mechanical inspection. Meter assemblies shall be inspected to verify that the materials, design, construction, physical dimensions, weight, marking, and workmanship are in accordance with the requirements of this specification.

4.6.2 Lifting. The meter assembly, ballasted to an evenly distributed gross weight of 500 lb minimum, shall be hoisted so that the entire weight of the meter assembly and the ballast are supported by the two lifting eyes for a minimum of 10 minutes. During and after the test, the entire meter assembly, especially the lifting eyes and the frame, shall be inspected for evidence of permanent deformation or failure. After this test, the meter assembly shall be subjected to the tow test in 4.6.5.

*4.6.3 Hydrostatic pressure test. Meter assemblies shall be hydrostatically tested as follows:

(a) With the caps on the outlet fittings, the two outlet valves open, and the inlet valve open, connect the meter assembly inlet coupling to a source of test fluid by means of suitable fittings.

(b) Fill the meter assembly with test fluid, being careful to eliminate all entrapped air.

(c) Raise inlet pressure to 200 psi and maintain this pressure for a minimum of 5 minutes.

(d) During the test, observe the meter assembly for leakage and evidence of failure.

(e) Repeat (a) through (d) except that the pressure shall be applied through one of the two outlets with the inlet valve open and the plug in the inlet fitting.

*4.6.4 Accuracy. Meter assemblies shall be tested for accuracy as follows:

(a) Incline any horizontal plane of the meter assembly to a minimum of 5° (in any direction) from any convenient true horizontal reference plane. The meter assembly shall remain in this attitude throughout the following tests.

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(b) Using a known volume of test fluid (see 4.2.2) of any magnitude sufficient to provide reliable readings, calibrate the meter by means of the micrometer adjuster (see 3.3.3.1) so that the registered volume (see 3.3.3) is within 0.02 percent of the true volume.

(c) After calibration, the following tests shall be made:

(1) With an input pressure of 85 psi and any discharge back pressure still permitting the required flow, the input valve shall be manipulated to obtain the desired rate of flow. Under these conditions, an accuracy test shall be made at each of the following flows: 15 gpm, 125 gpm, 250 gpm, and a maximum of 350 gpm.

(2) Repeat (1), except that the input pressure shall be 150 psi minimum.

(d) Record the micrometer adjuster setting, then change the setting to the extreme limit in the zero direction, and perform an accuracy test at an input pressure of 85 psi and a flow rate of 350 gpm. The register shall show an error within 0.04 percent of the deliberate error fed into the adjuster.

(e) Repeat (d), above, except that the micrometer adjuster shall be set to the opposite extreme.

*4.6.5 Towing. With the meter assembly completely filled with test fluid (see 4.2.2), the dust caps and plug in place, and all gate valves open, the meter assembly shall be towed from either end (selected at random) by means of the towing eyes, for a minimum distance of 400 feet across a concrete pavement at a speed of 5 to 7 miles per hour (mph). After towing, the meter assembly shall be subjected to the following inspection and tests:

(a) Visual inspection for major damage (nicks, dents, scrapes and marred finishes are not to be considered major damage).

(b) Measure fluid volume required (if any) to refill meter assembly to the original volume contained therein before the test.

(c) Hydrostatic pressure as specified in 4.6.3.

(d) Accuracy as specified in 4.6.4.

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*4.7 Preparation for delivery tests. A pilot pack shall be submitted to the following tests for conformance to the requirements of this specification. The pack shall contain the assembled meter and accessories secured in place and ready for shipment.

4.7.1 Impact test. The crate, suspended by means of slings or grabhook, shall be raised to a height sufficient to allow the crate to swing clear of the ground. The crate shall be swung to cause one end or corner to strike against a solid barrier at a speed of at least 8 mph. The test shall be repeated for the opposite end of the crate.

4.7.2 Drop test. One end of the crate shall be mounted on a wood sill a minimum of 6 inches high placed at right angles to the skids. The opposite end shall then be raised to a height of 36 inches and allowed to fall free to a hard surface. This test shall be applied once to each end.

4.7.3 Examination. After completion of the above tests, the pack shall be examined for deficiencies, which shall be corrected by redesign and the crate retested.

5. PREPARATION FOR DELIVERY

*5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

*5.1.1 Level A. After completion of acceptance testing, the meter shall be completely filled with preservative oil, conforming to grade 2 of MIL-L-21260. Oil shall be retained in the meter for one hour, after which the meter shall be completely drained and the dust caps and plug installed. Valves shall be open a minimum of one-quarter of an inch. Valves and end fittings shall be wrapped with barrier material conforming to grade C of MIL-B-121 and sealed with pressure sensitive tape conforming to PPP-T-60.

*5.1.2 Level C. The meter shall be preserved and packaged in accordance with the manufacturer's standard practice to afford protection against deterioration and damage.

*5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

*5.2.1 Level A. Each meter shall be packed in a container conforming to class 2 of PPP-B-621, or overseas type, style A or B, plywood type III, class 1 of PPP-B-601. Containers

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shall be modified by the addition of not less than two nominal 2-inch by 4-inch wood skids positioned flatwise across the bottom of the box. The ends of the skids shall be beveled full depth at a 45° angle. Blocking, bracing and anchoring of the meter within the container shall be in accordance with MIL-STD-1186.

*5.2.2 Level B. Each meter shall be packed in a container conforming to class 1 of PPP-B-621 or domestic type, style A or B, plywood type III, class 1 of PPP-B-601. Containers shall be modified by the addition of skids as indicated for level A. Blocking, bracing and anchoring of the meter within the container shall be in accordance with MIL-STD-1186.

*5.2.3 Level C. Each meter assembly shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with the carrier rules and regulations applicable to the mode of transportation.

5.3 Marking. Shipments shall be marked in accordance with MIL-STD-129.

6. NOTES

*6.1 Intended use. The meter assemblies of this specification form a portion of an amphibious assault fuel system, and are intended to meter fuels conforming to VV-F-800 (fuel oil, diesel), MIL-G-3056 (automotive combat gasoline), MIL-G-5572 (aviation gasoline), and MIL-J-5624 (jet fuels, JP-4 and JP-5).

*6.2 Ordering data. Procurement documents should include the following:

(a) Title, number, and date of this specification.

(b) When first article is not required.

(c) Levels of preservation, packaging and packing (see 5.1 and 5.2).

*6.3 Color chips. Color chips matching Marine Corps green, No. 23, lusterless, may be obtained from the Commandant of the Marine Corps (Code CSY-10) Headquarters, U. S. Marine Corps, Washington, D. C. 20380.

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*6.4 First article instructions. The first article shall be examined and tested for approval at the contractor's plant or at an independent commercial testing laboratory acceptable to the procuring activity. First article tests shall be witnessed by a Government representative of the procuring activity. Approval of the first article shall be by the contracting officer.

*6.5 Marginal notations. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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